Jing Xu

Curriculum Vitae

Cybersecurity Group, Delft University of Technology,
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The Netherlands
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Born: 4 April, 1994
Nationality: Chinese
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Education

October 2019–now **PhD candidate in Computer Science**, *Delft University of Technology*, Delft, The Netherlands, Study in the field of *GNNs, Security of AI, Explainable AI*
Supervised by Stjepan Picek. Planned to finish in Dec. 2023.

Sep. 2016–Jun. 2019 **MSc in Optical Engineering**, *Beihang University*, Beijing, China, *Thesis: Research on Multi-Source Information Fusion in the All-Source Navigation and Positioning System based on the Factor Graph*
GPA: 3.857/4.0-RANK: top 5%

Sep. 2012–Jul. 2016 **BSc in Electrical Engineering**, *Shanghai University*, Shanghai, China, *Thesis: Coin Automatic Recognition System based on Computer Vision*
GPA:3.86/4.0-RANK: 1/931

Experience

Mar. 2019–Sep. 2019 **Research Intern**, *Momo Technology Company Limited, Deep Learning Lab*, Beijing, China

- Research on face recognition, object detection methods.
- Apply GANs to anti-spoofing face recognition task.

Research Interests

- Graph Neural Networks
- Security of AI, e.g., backdoor attacks, membership inference attacks
- Explainable AI

Publications

2022 **"More is Better (Mostly): On the Backdoor Attacks in Federated Graph Neural Networks." Jing Xu***, Rui Wang, Kaitai Liang, Stjepan Picek. Annual Computer Security Applications Conference (ACSAC).

2022 **"Poster: Clean-label Backdoor Attack on Graph Neural Networks." Jing Xu***, Stjepan Picek. CCS.

2022 **"Label-Only Membership Inference Attack against Node-Level Graph Neural Networks."** Mauro Conti, Jiaxin Li*, Stjepan Picek, **Jing Xu**. AISec, CCS Workshop.

2022 **"Can You Hear It? Backdoor Attacks via Ultrasonic Triggers."** Stefanos Koffas, **Jing Xu**, Mauro Conti, Stjepan Picek. The ACM Workshop on Wireless Security and Machine Learning (WiseML).

2022 **"Watermarking Graph Neural Networks based on Backdoor Attacks." Jing Xu**, Stjepan Picek. Under revision for Euro S&P.

2021 **"Explainability-based backdoor attacks against graph neural networks." Jing Xu**, Minhui(Jason) Xue, Stjepan Picek. The ACM Workshop on Wireless Security and Machine Learning (WiseML).

2021 **"A multi-sensor information fusion method based on factor graph for integrated navigation system." Jing Xu**, Gongliu Yang, Yiding Sun, Stjepan Picek. IEEE Access, 9, pp. 12044-12054.

2017 **"Coin Recognition Method Based on SIFT Algorithm." Jing Xu**, Gongliu Yang, Yuanyuan Liu, Jingjia Zhong. 4th International Conference on Information Science and Control Engineering (ICISCE).

2017 **"Initial alignment of strapdown inertial navigation system with the aid of Doppler velocity log."** Jingjia Zhong, Gongliu Yang, Yuanyuan Liu, **Jing Xu**. ICISCE.

Computer skills

Languages:	Python, C, MATLAB	Frameworks:	PyTorch, TensorFlow
Operation Systems:	Linux, Windows	Developer Tools:	VS Code, Git, Slurm

Honors

Mar. 2018–BUAA Outstanding Graduate Student, Outstanding Member

Oct. 2017–BUAA The First Prize Scholarship (for two consecutive years)

Jun. 2016–SHU Outstanding Graduate Student, Outstanding Student, Outstanding Member

Nov. 2015–SHU The First Prize Scholarship (for three consecutive years), The GuangHua Scholarship

Selected Research Visits and Collaborations

Mar. 2022 **Visiting researcher**, *Padova University*, Padova, Italy

Nov. 2021 **Visiting researcher**, *TU Darmstadt*, Darmstadt, Germany

Teaching and Supervision

Course

2022– Security and Privacy of Machine Learning, Radboud University, Netherlands.

Supervision

Master Student Alex Simonov, 2022–, Delft University of Technology.